

Question 1

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Determine if each set below is convex.

1. $\{(x, y) \in \mathbb{R}_{++}^2 \mid x/y \leq 1\}$

since $(x, y) \in \mathbb{R}_{++}^2 \Rightarrow 0 < x \leq y$

Let $H = \{(x, y) \in \mathbb{R}_{++}^2 \mid x/y \leq 1\}$

$= \{(x, y) \mid x - y \leq 0\} \cap \mathbb{R}_{++}^2$

$\Rightarrow$ Intersection of halfspaces are convex

2. $\{(x, y) \in \mathbb{R}_{++}^2 \mid x/y \geq 1\}$

Since $(x, y) \in \mathbb{R}_{++}^2 \Rightarrow 0 < y \leq x$

$\Rightarrow \{(x, y) \in \mathbb{R}_{++}^2 \mid x/y \geq 1\} = \{(x, y) \mid y - x \leq 0\} \cap \mathbb{R}_{++}^2$

$\Rightarrow$ Intersection of halfspaces are convex

3. $\{(x, y) \in \mathbb{R}_{++}^2 \mid xy \geq 1\}$

Since $(x, y) \in \mathbb{R}_{++}^2 \Rightarrow xy \geq 1 \Rightarrow y \geq \frac{1}{x}$

$\Rightarrow \{(x, y) \in \mathbb{R}_{++}^2 \mid xy \geq 1\} = \{(x, y) \mid y \geq \frac{1}{x}\} \cap \mathbb{R}_{++}^2 \quad (x > 0)$

$y = f(x) = \frac{1}{x}$ is convex $\Rightarrow \text{epi}(f(x))$ is convex

$\Rightarrow$ Intersection is convex

4. $\{(x, y) \in \mathbb{R}_{++}^2 \mid xy \leq 1\}$

Not convex. e.g. when $p_1 = (2, \frac{1}{2})$, $p_2 = (\frac{1}{2}, 2)$

$\Rightarrow \bar{p} = (\frac{1}{2}(2 + \frac{1}{2}), \frac{1}{2}(2 + \frac{1}{2})) = (\frac{5}{4}, \frac{5}{4}) \notin \{(x, y) \in \mathbb{R}_{++}^2 \mid xy \leq 1\}$

5. $S = \{x \in \mathbb{R}^n \mid x^T y \leq 1 \text{ for all } y \in C\}$, where C is a set (may not be convex).

$S = \bigcap_{y \in C} \{x \in \mathbb{R}^n \mid x^T y \leq 1\} \Rightarrow$ each is a halfspace, intersection is convex.

6. The ellipsoid $\{x | x^T P^{-1} x \leq 1\}$ where $P \in S_{++}^n$.

since $P \in S_{++}^n \Rightarrow P^{-1} \geq 0$, let $A = P^{-1}$

let $f(x) = x^T A x \Rightarrow f'(x) = Ax + x^T A = 2Ax \Rightarrow f''(x) = 2A \geq 0$

$\Rightarrow$ convex, and sublevel set is convex.

Question 2

Give an example of two closed convex sets that are disjoint but cannot be strictly separated.

let $C_1 = \{(x, y) | y \geq e^x\}$, $C_2 = \{(x, y) | y \leq 0\}$

both are convex and closed.

$\lim_{x \rightarrow -\infty} e^x - 0 = 0 \Rightarrow$ no positive margin

$\Rightarrow$ cannot be strictly separated.

Question 3

Supporting hyperplanes.

1. Express the closed convex set $\{x \in \mathbb{R}_+^2 | x_1 x_2 \geq 1\}$ as an intersection of halfspaces.

let $C = \{x \in \mathbb{R}_+^2 | x_1 x_2 \geq 1\}$, $f(x) = \frac{1}{x}$, ($x > 0$)

$\Rightarrow C = \{(x, y) | y \geq \frac{1}{x}, x > 0, y > 0\}$

for $\forall x_0 \in (0, +\infty)$, $f(x_0) = \frac{1}{x_0}$.

$f'(x) = -\frac{1}{x^2}$. $\Rightarrow$ support hyperplane $\Rightarrow y - \frac{1}{x_0} = -\frac{1}{x_0^2}(x - x_0)$
 $\Rightarrow y = -\frac{1}{x_0^2}x + \frac{2}{x_0}$

$$\Rightarrow C = \bigcap_{x_0 \in (0, +\infty)} \left\{ (x, y) \mid y \geq -\frac{1}{x^2}x + \frac{2}{x_0} \right\}.$$

these are all halfspaces.

2. Let $C = \{x \in \mathbb{R}^n \mid \|x\|_\infty \leq 1\}$, the l_∞ -norm unit ball in $\mathbb{R}^n$, and let $\hat{x}$ be a point in the boundary of C . Identify the supporting hyperplanes of C at $\hat{x}$ explicitly.

$$\hat{x} \in \text{bd}(C) \Rightarrow \max(\hat{x}_1, \dots, \hat{x}_n) = 1$$

$$\Rightarrow \text{for each } i \in \{1, \dots, n\}, \quad x_i = 1 \text{ if } \hat{x}_i = 1, \\ x_i = -1 \text{ if } \hat{x}_i = -1$$

3. Express the ball $B = \{x \in \mathbb{R}^n \mid \|x\|_2 \leq 1\}$ as an intersection of halfspaces.

$$\text{let } f(u) = u^T x \leq 1, \quad \|u\|_2 = 1.$$

$$\text{if } \|x\|_2 \leq 1, \text{ then } u^T x \leq \|u\|_2 \cdot \|x\|_2 \leq 1 \text{ for } \forall \|u\|_2 = 1$$

$$\text{if } u^T x \leq 1 \text{ for } \forall \|u\|_2 = 1, \text{ let } u = \frac{x}{\|x\|_2}, \quad \|x\|_2 = u^T x \leq 1.$$

$$\Rightarrow B = \{x \in \mathbb{R}^n \mid \|x\|_2 \leq 1\} = \bigcap_{\|u\|_2 = 1} \{x \mid u^T x \leq 1\}.$$

Question 4

Perspective function $P : \mathbb{R}^{n+1} \rightarrow \mathbb{R}^n$ is defined as:

$$P(x, t) = x/t, \quad \text{dom } P = \{(x, t) \mid t > 0\}$$

where $x \in \mathbb{R}^n, t > 0$ (i.e. $\text{dom}(P) = \{(x, t) \mid x \in \mathbb{R}^n, t > 0\}$.) Prove:

- If $C \subset \text{dom}(P)$ is convex, then $P(C)$ is convex.

$$\forall p_1 = (x_1, t_1), p_2 = (x_2, t_2) \in C, \quad \theta \in [0, 1]$$

Since C is convex $\hat{p} = \theta p_1 + (1 - \theta) p_2$ should always be in C

$$\hat{p} = (\theta \cdot x_1 + (1-\theta) \cdot x_2, \theta t_1 + (1-\theta) t_2)$$

$$\Rightarrow P(\hat{p}) = \frac{\theta x_1 + (1-\theta) \cdot x_2}{\theta t_1 + (1-\theta) t_2}$$

$$P(p_1) = \frac{x_1}{t_1}, \quad P(p_2) = \frac{x_2}{t_2}, \quad \text{let } \lambda = \frac{\theta t_1}{\theta t_1 + (1-\theta) t_2}$$

$$\lambda \frac{x_1}{t_1} + (1-\lambda) \frac{x_2}{t_2} = \frac{\theta t_1 x_1}{t_1 \cdot (\theta t_1 + (1-\theta) t_2)} + \frac{(1-\theta) t_2 x_2}{t_2 (\theta t_1 + (1-\theta) t_2)}$$

$$= \frac{\theta x_1 + (1-\theta) x_2}{\theta t_1 + (1-\theta) t_2} = P(\hat{p})$$

$\Rightarrow$ any combination of $P(p_1)$ and $P(p_2)$, i.e. $P(\hat{p})$, still on $P(C)$
 $\Rightarrow$ convex.

• If $D \subset \mathbb{R}^n$ is convex, then $P^{-1}(D)$ is convex.

$$P^{-1}(D) = \{(x, t) \mid t > 0, x/t \in D\}.$$

$$\forall (x_1, t_1), (x_2, t_2) \in P^{-1}(D), \text{ let } y_1 = \frac{x_1}{t_1} \in D, y_2 = \frac{x_2}{t_2} \in D$$

$$\text{let } \theta \in [0, 1], \text{ let } t = \theta t_1 + (1-\theta) t_2 > 0, \quad y = \frac{\theta t_1}{t} y_1 + \frac{(1-\theta) t_2}{t} y_2$$

$$\theta x_1 + (1-\theta) x_2 = \theta t_1 y_1 + (1-\theta) t_2 y_2 = t y$$

$$\Rightarrow (\theta x_1 + (1-\theta) x_2, \theta t_1 + (1-\theta) t_2) = (t y, t) \in P^{-1}(D)$$

$$\Rightarrow P^{-1}(D) \text{ is convex}$$